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Randomized approaches to approximating solutions of large kernel systems for Kernel Ridge Regression using the Nyström method

Bachelor's Project Mathematics

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Abstract

Kernel ridge regression leads to a linear system involving the full kernel matrix K. This matrix has size $n \times n$, so forming and storing it becomes expensive when the number of data points is large. This project studies randomized Nyström approximations, where only $l \ll n$ columns of K are used to build a low-rank approximation. The aim is to compare how the choice of sampled columns and the number of sampled columns affect accuracy, runtime, memory use, conditioning, and solver behaviour. (Gittens and Mahoney, 2013)

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1 Introduction

Kernel methods provide a way to capture nonlinear relationships by projecting data onto a higher-dimensional space and describing the data through pairwise similarities. Kernel functions are used to compute inner products in the higher-dimensional space without explicitly mapping the data. This allows flexible regression models to be constructed, but the resulting matrix computations become expensive as the number of observations increases (Alaoui and Mahoney, 2015). This project therefore studies randomized Nyström approximation, one of the most popular techniques that reduces this computational cost for kernel ridge regression while trying to preserve its prediction quality (Musco and Musco, 2017).

Many regression problems start with data (x_i, y_i) for $i = 1, \ldots, n$, where n is the number of observations, $x_i \in \mathbb{R}^d$ are the inputs and $y_i \in \mathbb{R}$ are the outputs. We would like to use these data to create a model for the relationship between the inputs and outputs. One formulation of this task is kernel ridge regression, which uses a kernel function k to measure pairwise similarities between the inputs. These values form the $n \times n$ kernel matrix K, whose entries are $K_{ij} = k(x_i, x_j)$. The details of this formulation are introduced in Chapter 2.

For large datasets, the main computational difficulty is the need to explicitly form the full matrix K, which costs $O(n^2)$ time and memory. This is the part that becomes infeasible first and takes the most time and resources. Solving the resulting system directly can also cost about $O(n^3)$ operations, so this is fine for small tests, but not realistic when n becomes large (Drineas et al., 2005; Alaoui and Mahoney, 2015).

Therefore in this project we investigate the Nyström approximation of the kernel matrix. Instead of forming all of K, we select an index set $I \subset \{1, \ldots, n\}$ containing $l \ll n$ column indices. The selected columns form the matrix $C = K_{:,I} \in \mathbb{R}^{n \times l}$, while their intersections with the corresponding rows form the smaller matrix $W = K_{I,I} \in \mathbb{R}^{l \times l}$. These two matrices are then used to construct a low-rank approximation $\tilde{K} \approx K$, for example $\tilde{K} = CW^{\dagger}C^T$. This is a randomized approach for kernel matrices (Drineas et al., 2005; Gittens and Mahoney, 2013). The point is to reduce the computational cost while still keeping enough information from the original kernel matrix. The full construction is introduced in Chapter 3.

Nyström needs a set of l columns to build $\tilde{K}$. The simplest choice is uniform random sampling, i.e. pick columns uniformly at random. Smarter strategies try to pick columns that are more informative so that fewer columns are needed for similar accuracy. One idea is to sample according to ridge leverage scores, which give higher probability to points that matter more for the kernel ridge regression solution (Alaoui and Mahoney, 2015). Another idea is to approximate these leverage type probabilities adaptively without ever constructing the whole K (Calandriello et al., 2017).

This project is therefore about how to decide which columns to sample and how many columns l are needed for a good Nyström approximation in the KRR setting. We will compare different sampling strategies, starting from uniform random sampling as a baseline, and then compare against smarter strategies motivated by the literature. The experiments will compare these strategies in terms of prediction accuracy and computational complexity, such as runtime and memory, as l changes.

The research questions are:

- What Nyström column sampling strategies give good performance for KRR in terms of prediction accuracy vs runtime/memory?
- How large does l need to be to get a good approximation, and how does this depend

on the sampling strategy and λ?

- What do the accuracy and cost (runtime/memory) trade-offs look like when l increases, and can we recommend a reasonable approach?
- How do conditioning and solver convergence relate to λ and the Nyström approximation quality?

This thesis is organised as follows. Chapter 2 introduces kernel ridge regression and the kernel matrices used in the project. It also gives the first singular value experiments, which motivate why a low-rank approximation is a reasonable thing to try. Chapter 3 introduces the Nyström approximation, starting from the general positive semidefinite matrix formula and then moving to column sampling.

2 Kernel Ridge Regression

In this chapter we introduce the kernel ridge regression problem that lays the foundation for this project. First, the regression problem is written as an optimization problem over a reproducing kernel Hilbert space. Then the representer theorem is used to turn this function-space problem into a finite-dimensional problem involving the kernel matrix. Finally, we introduce the kernels used in the experiments and give some numerical checks that motivate using low-rank approximations.

2.1 Introduction to the regression problem

In this section we introduce the regression problem that will be used throughout the project. Assume we have a dataset (x_i, y_i) for $i = 1, \ldots, n$, where $x_i \in \mathbb{R}^d$ are inputs and $y_i \in \mathbb{R}$ are outputs. We would like to use these data to create a model for the relationship between the inputs and outputs.

In ordinary linear regression, the model is searched for in a finite-dimensional linear function class. Kernel ridge regression instead searches over a reproducing kernel Hilbert space $\mathcal{H}$. Kernel ridge regression searches for a function $f \in \mathcal{H}$ that fits the training data while keeping the RKHS norm small. The optimization problem is

$$\min_{f \in \mathcal{H}} \|f(X) - y\|_2^2 + \lambda \|f\|_{\mathcal{H}}^2,$$

which is also

$$\min_{f \in \mathcal{H}} \sum_{i=1}^{n} (f(x_i) - y_i)^2 + \lambda \|f\|_{\mathcal{H}}^2.$$

Here $\lambda > 0$ is the regularization parameter. The first term measures the error on the training data, while the second term penalizes functions with large RKHS norm. In other words, λ controls the trade off between fitting the data and overfitting.

The optimization problem above is still written over a function space, thus in the next sections we will define the RKHS and discuss its connection to kernels.

2.1.1 Kernel trick

Suppose the data matrix is

$$X \in \mathbb{R}^{n \times d},$$

where each row is an input point $x_i \in \mathbb{R}^d$. A kernel method can be viewed as first mapping the data onto a feature space by a map

$$\phi : \mathcal{X} \to \mathcal{F}.$$

If the feature vectors were written as rows of a matrix, this would give an implicit feature matrix

$$\Phi \in \mathbb{R}^{n \times d'},$$

where d' may be much larger than d or even infinite. The corresponding kernel matrix is

$$K = \Phi \Phi^T \in \mathbb{R}^{n \times n},$$

with entries

$$K_{ij} = k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle_{\mathcal{F}}.$$

The kernel trick is that we do not need to explicitly construct $\phi(x_i)$ or the matrix Φ. Instead, we compute the inner products in the feature space directly through the kernel function $k(x_i, x_j)$.

2.1.2 Positive definite kernels and RKHS

Before defining the RKHS, we first need to specify what kind of kernel functions can be used. The reason is that the kernel will later be used to define an inner product between functions, so it must satisfy the basic properties needed for an inner product to be valid.

Definition 2.1 (Positive definite kernel)**.** Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the input space. A symmetric function

$$k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$$

is called a positive definite kernel if, for any points $x_1, \ldots, x_n \in \mathcal{X}$ and any coefficients $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$,

$$\sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j k(x_i, x_j) \geq 0.$$

The symmetry condition means that $k(x_i, x_j) = k(x_j, x_i)$, which matches the symmetry of an inner product. The positive definiteness condition will be used below to make sure that the squared norm induced by the kernel-defined inner product is nonnegative.

The value $k(x_i, x_j)$ measures the similarity between two inputs x_i and x_j according to the chosen kernel. If one input is fixed then $k(\cdot, x_i)$ is a function of the remaining argument. This is called a kernel section centered at x_i.

The RKHS associated with k is built from these kernel sections. First consider finite linear combinations of kernel sections with arbitrary centers in the input space, not only the training points.

$$f(\cdot) = \sum_{i=1}^{n} \alpha_i k(\cdot, x_i), \qquad x_i \in \mathcal{X}.$$

On this space, the inner product is first defined on individual kernel sections by

$$\langle k(\cdot, x_i), k(\cdot, x_j) \rangle_{\mathcal{H}} = k(x_i, x_j).$$

The inner product rule is then extended linearly from individual kernel sections to finite linear combinations. Therefore, for two finite linear combinations

$$f(\cdot) = \sum_{i=1}^{m} \alpha_i k(\cdot, x_i), \qquad g(\cdot) = \sum_{j=1}^{r} \beta_j k(\cdot, y_j),$$

the inner product is

$$\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^{m} \sum_{j=1}^{r} \alpha_i \beta_j k(x_i, y_j).$$

In particular, when $g = f$, this gives the RKHS norm. We have

$$\|f\|_{\mathcal{H}}^2 = \left\langle \sum_{i=1}^{n} \alpha_i k(\cdot, x_i), \sum_{j=1}^{n} \alpha_j k(\cdot, x_j) \right\rangle_{\mathcal{H}} = \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j \langle k(\cdot, x_i), k(\cdot, x_j) \rangle_{\mathcal{H}}.$$

by the bilinearity of the inner product. Then we use the reproducing property to get

$$\|f\|_{\mathcal{H}}^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j k(x_i, x_j).$$

To ensure that the squared norm is nonnegative, we therefore need the kernel to be positive definite. This is the pre-Hilbert space generated by the kernel sections, then we can obtain the RKHS by completing this space (Zhu, 2023).

Definition 2.2 (Reproducing kernel Hilbert space)**.** Given a positive definite kernel k, the associated reproducing kernel Hilbert space $\mathcal{H}$ is the Hilbert space obtained by completing the span of the kernel sections:

$$\mathcal{H} = \overline{\text{span}\{k(\cdot, x) : x \in \mathcal{X}\}}.$$

It has the reproducing property

$$f(x) = \langle f, k(\cdot, x)\rangle_{\mathcal{H}}$$

for every $f \in \mathcal{H}$ and every $x \in \mathcal{X}$.

The reproducing property means that evaluating f at a point can be written as an inner product with the kernel section centered at that point (Zhu, 2023).

The RKHS norm $\|f\|_{\mathcal{H}}$ gives a way to penalize the size of the function in this space. In later sections, after the representer theorem shows that the KRR solution has the form

$$f(\cdot) = \sum_{i=1}^{n} \alpha_i k(\cdot, x_i),$$

the same norm formula becomes

$$\|f\|_{\mathcal{H}}^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j k(x_i, x_j) = \alpha^T K \alpha.$$

Thus the positive definiteness condition will also imply that the kernel matrix K is positive semidefinite.

One reason this norm can be interpreted as controlling the variation of f is the Cauchy-Schwarz inequality together with the reproducing property. For two inputs x and x',

$$|f(x) - f(x')| = |\langle f, k(\cdot, x) - k(\cdot, x')\rangle_{\mathcal{H}}| \leq \|f\|_{\mathcal{H}} \|k(\cdot, x) - k(\cdot, x')\|_{\mathcal{H}}.$$

Thus a smaller norm limits how much the function can vary relative to the distance induced by the kernel. This motivates the regularization term $\lambda\|f\|_{\mathcal{H}}^2$ in kernel ridge regression (Zhu, 2023).

The KRR objective is now well defined, but we still have to deal with the infinite dimensionality of $\mathcal{H}$. We will discuss this in the next sections.

2.1.3 Representer form and the kernel matrix

The optimization problem above is written over the whole function space $\mathcal{H}$, so at first it is not clear how it can be solved computationally. The representer theorem gives the key reduction (Zhu, 2023).

Theorem 2.1 (Representer theorem)**.** *Let $\{(x_i, y_i)\}_{i=1}^n$ be a training set, and let $\mathcal{H}$ be the RKHS associated with the kernel k. Consider an optimization problem of the form*

$$\hat{f} = \arg\min_{f \in \mathcal{H}} \mathcal{L}\big(\{y_i, f(x_i)\}_{i=1}^n\big) + p\big(\|f\|_{\mathcal{H}}^2\big),$$

where $\mathcal{L}$ is a loss function depending only on the fitted values at the training points and p is a monotone penalty function. Then the minimizer can be written as a finite linear combination of kernel functions centered at the training points:

$$\hat{f}(\cdot) = \sum_{i=1}^{n} \alpha_i k(\cdot, x_i),$$

for some coefficients $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$.

For KRR, this theorem applies because the loss depends only on $f(x_1), \ldots, f(x_n)$ and the penalty is $\lambda \|f\|_{\mathcal{H}}^2$. The reason why this is useful is that we no longer need to search over all functions in $\mathcal{H}$. We have reduced the problem from infinite dimensional to finite dimensional, then we only need to determine the n coefficients $\alpha_1, \ldots, \alpha_n$. To see why this is possible, let

$$\mathcal{S} = \text{span}\{k(\cdot, x_1), \ldots, k(\cdot, x_n)\}.$$

Any function $f \in \mathcal{H}$ can be decomposed into a part inside this span and a part orthogonal to it, so

$$f = f_{\mathcal{S}} + h, \qquad f_{\mathcal{S}} \in \mathcal{S}, \qquad h \perp \mathcal{S}.$$

For each training point x_i, the reproducing property gives

$$h(x_i) = \langle h, k(\cdot, x_i) \rangle_{\mathcal{H}} = 0.$$

Therefore

$$f(x_i) = f_{\mathcal{S}}(x_i) + h(x_i) = f_{\mathcal{S}}(x_i),$$

so replacing f by $f_{\mathcal{S}}$ does not change any fitted value on the training data. This means

$$\sum_{i=1}^{n} (f(x_i) - y_i)^2 = \sum_{i=1}^{n} (f_{\mathcal{S}}(x_i) - y_i)^2.$$

At the same time,

$$\|f\|_{\mathcal{H}}^2 = \|f_{\mathcal{S}}\|_{\mathcal{H}}^2 + \|h\|_{\mathcal{H}}^2,$$

so the extra part can only increase the penalty term. Hence the minimizer can be taken inside the finite-dimensional span $\mathcal{S}$ (Zhu, 2023).

To rewrite the objective in matrix form, we will define the kernel matrix

$$K \in \mathbb{R}^{n \times n}, \qquad K_{ij} = k(x_i, x_j).$$

Since k is symmetric and positive definite, the matrix K is symmetric positive semidefinite.

We now rewrite both terms in the KRR objective using α and K. For the training points we have

$$f(x_i) = \sum_{j=1}^{n} \alpha_j k(x_i, x_j),$$

so in vector form

$$f(X) = K\alpha.$$

Therefore the data fitting term becomes

$$\sum_{i=1}^{n} (f(x_i) - y_i)^2 = \|K\alpha - y\|_2^2.$$

The RKHS norm term can also be written using the kernel matrix. Since

$$\|f\|_{\mathcal{H}}^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j k(x_i, x_j)$$

. We use $K_{ij} = k(x_i, x_j)$ and get

$$\|f\|_{\mathcal{H}}^2 = \alpha^T K \alpha.$$

Substituting the data fitting term and the RKHS norm term into the objective gives the finite-dimensional problem

$$\min_{\alpha \in \mathbb{R}^n} \|K\alpha - y\|_2^2 + \lambda \alpha^T K \alpha.$$

Expand

$$\|K\alpha - y\|_2^2 = (K\alpha - y)^T(K\alpha - y).$$

Then differentiate

$$(K\alpha - y)^T(K\alpha - y) + \lambda\alpha^T K\alpha$$

with respect to α. Since K is symmetric this gives

$$2K(K\alpha - y) + 2\lambda K\alpha = 0.$$

This then reduces to

$$(K + \lambda I)\alpha = y.$$

Therefore the coefficient vector is

$$\alpha = (K + \lambda I)^{-1}y.$$

The prediction for a new point x is then

$$\hat{f}(x) = \sum_{j=1}^{n} \alpha_j k(x, x_j).$$

If X_{test} is a set of test points and $K_{\text{test,train}}$ is the matrix with entries

$$(K_{\text{test,train}})_{ij} = k(x_i^{\text{test}}, x_j^{\text{train}}),$$

then the test predictions are

$$\hat{y}_{\text{test}} = K_{\text{test,train}}\alpha.$$

This completes the main transition in this chapter, the original problem over functions in the space $\mathcal{H}$ has been reduced to solving a finite-dimensional linear system with kernel matrix K.

2.2 Computational cost of full KRR

The regularization term is important numerically. Since K can have very small eigenvalues or even zero eigenvalues, adding λI shifts all eigenvalues by λ so $K + \lambda I$ is positive definite. This makes the system uniquely solvable and improves the conditioning of the problem.

For large datasets, the main computational difficulty is that K is an $n \times n$ matrix. Forming the full kernel matrix requires $O(n^2)$ kernel evaluations and storing it requires $O(n^2)$ memory. Solving the system directly, for example by Cholesky decomposition, costs $O(n^3)$ operations. For small n this is fine but for large datasets this quickly becomes infeasible.

This is the computational reason for studying kernel approximation. The matrix formulation is finite-dimensional, but the dimension is the number of training observations n. Therefore when n is large it is natural to replace the full kernel matrix by a cheaper approximation $K \approx \widetilde{K}$.

The rest of this chapter studies the behaviour of the full kernel matrix and the full KRR solution, then we introduce the Nyström approximation in Chapter 3.

2.3 Kernel choices

In the experiments we start with three common kernels. The linear kernel is

$$k(x, z) = x^T z.$$

The polynomial kernel used in the first tests is

$$k(x, z) = (x^T z + 1)^2.$$

The Gaussian radial basis function kernel (rbf) is

$$k(x, z) = \exp(-\gamma \|x - z\|_2^2).$$

The parameter γ controls how local the Gaussian kernel is (Zhu, 2023; scikit-learn developers, 2026). Larger γ makes points only strongly similar to very nearby points while smaller γ makes the similarities decay more slowly.

These choices give a range of model flexibility. The linear kernel is the simplest option and is used to capture linear structure in the data. The polynomial kernel can capture nonlinear feature interactions unlike the linear kernel. The RBF kernel is used as a more flexible nonlinear kernel, and its behaviour depends strongly on the choice of γ. Since this project is about KRR and Nyström approximation, these kernels are used mainly to compare different kernel matrices rather than to give a general rule for which kernel is best.

2.4 Singular values and low-rank structure

Given a large matrix, if many singular values are very small then much of the matrix can be approximated using a smaller rank. This is why in this section we look at singular values of K for real data. The point is to check whether a low-rank approximation is reasonable for real world data.

For this real data experiment we use the Superconductivity dataset from the UCI Machine Learning Repository (Hamidieh, 2018). This is a regression dataset with $n = 21263$ samples and $d = 81$ numerical input features, and the output is the critical temperature. This fits the KRR setting since the target variable is continuous.

This dataset is also large enough to show the computational problem. If the full dataset is used, the kernel matrix has size $n \times n$ where $n = 21,263$, this means there are $n^2 \approx 10^8$ entries in K which corresponds to about 3.6 GB in memory, not to mention the construction computational cost.

This is before solving the linear system or computing any decomposition. So the full matrix is already expensive, and this is the reason that an approximation method such as Nyström is relevant.

We now look at the singular values of three kernels linear, polynomial, and Gaussian RBF. The singular values are computed using the full kernel matrix, so this is just a check of the low-rank structure. The point is to see whether the singular values decay in practice for real data.

For the singular value plots, the singular values are normalised by the largest singular value, so the plotted quantity is

$$\frac{\sigma_i(K)}{\sigma_1(K)}.$$

This makes the three kernels easier to compare since all three curves start at 1. The plots are also stopped once the singular values reach around 10^{-16}, since after that the remaining tail is machine precision.

Figure 2.1 compares the three kernels used in this project. Figure 2.1(a) shows the linear kernel. Since $K = XX^T$ for a linear kernel and the data have 81 features, the rank is at most 81, so the singular values drop to zero after that point. Figure 2.1(b) shows the polynomial kernel. This kernel is still based on inner products, but the square introduces nonlinear factors, so its spectrum does not have the same clean cut-off as the linear kernel. Figure 2.1(c) shows the RBF kernel. Here the decay depends more on the geometry of the data and the value of γ, so this is the most interesting case out of all three kernels.

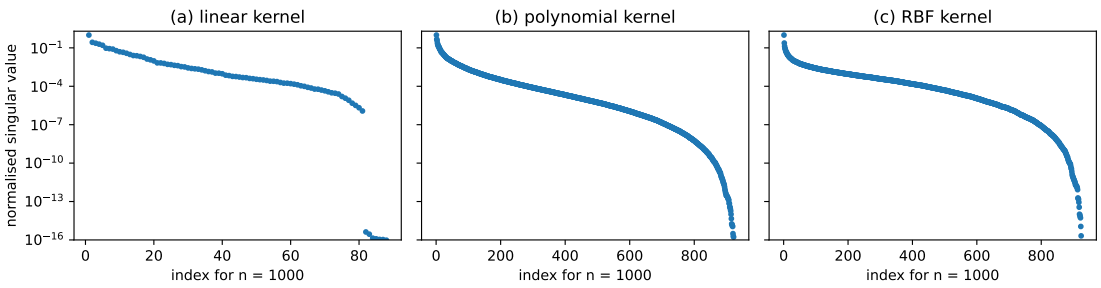


Figure 2.1: Normalised singular value decay for the three kernel matrices: (a) linear, (b) polynomial and (c) RBF.

The decay in these plots motivates replacing K by a lower-rank matrix $\widetilde{K}$ in later chapters. The goal is to keep the main spectral information needed for the KRR computation while reducing the cost.

2.5 Runtime and accuracy experiments

We now check the computational cost in this section. Constructing a full kernel matrix means evaluating pairwise similarities between observations so the expected cost is quadratic in n. Solving the full KRR has cubic asymptotic cost if we use Cholesky or some other direct solver. However in the tested range the construction of K can still take longer than the solve. This does not contradict the asymptotic costs. It just means the $O(n^2)$ construction curve can start much higher since every kernel entry has to be evaluated and stored while the solve is done by highly optimized linear algebra methods. In practice the construction is almost always more expensive than the solve.

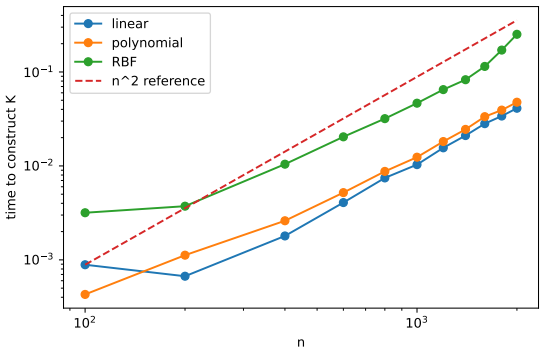


Figure 2.2: Time to construct the full kernel matrix for different values of n.

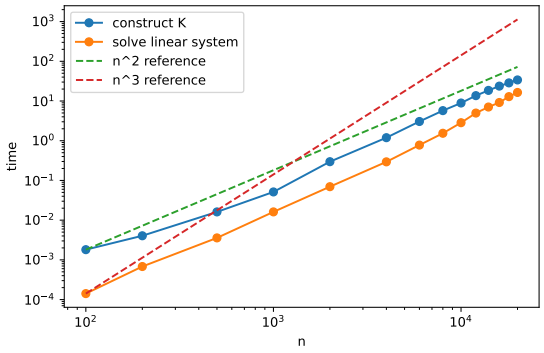


Figure 2.3: Time to construct K vs time to solve the full KRR system.

Figure 2.2 shows the construction time for the linear, polynomial and RBF kernels. The general behaviour follows the expected quadratic growth, but the constants are not the same. RBF kernel is slower since it is a more complex function than the other two.

Figure 2.3 compares the time to construct the RBF kernel matrix with the time to solve $(K + \lambda I)\alpha = y$. In the code, we measure the solve time by forming $K + \lambda I$, then applying a Cholesky factorization and two triangular solves. The construction time is larger for the increasing values of n which agrees with the theoretical expectation we have discussed above. This is one of the main reasons why column sampling is important: if only l columns are sampled, then we can build $K_{:,I}$ and $K_{I,I}$ instead of all of K which we will see in Chapter 3.

We now look at the behaviour of the full KRR solution as the regularization parameter λ changes. This is important because λ appears directly in the system $(K + \lambda I)\alpha = y$, so changing λ changes both the model and the conditioning of the linear system. Later, when K is replaced by an approximation $\tilde{K}$ in Nyström, the same parameter still appears in the approximation system. Therefore it is useful to first understand the effect of λ in the full KRR system.

For this experiment, we use a fixed train/test split and measure the test error using MSE. We used MSE since it is a natural metric for the regression error and it gives a simple baseline that later can be compared with the Nyström approximations.

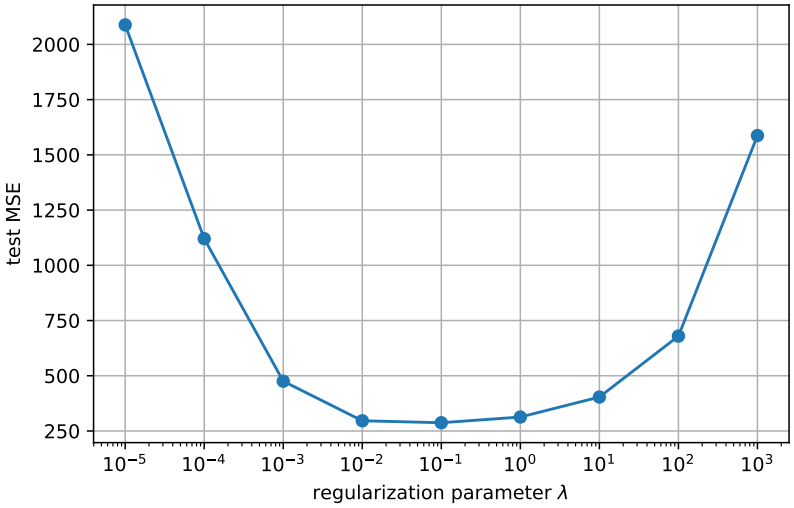


Figure 2.4: Test MSE for full KRR as λ changes.

Figure 2.4 shows that the test MSE depends strongly on λ. For very small values, especially $\lambda = 10^{-5}$ and $\lambda = 10^{-4}$, the test error is high. In this range the regularization term is too weak, so the solution can fit the training data too closely and generalize poorly. As λ increases, the test MSE drops sharply and reaches its lowest value around $\lambda = 10^{-1}$. After this point the error rises again, and for very large values such as $\lambda = 10^{2}$ and $\lambda = 10^{3}$, the penalty term is too strong and the fitted function becomes too restricted. Thus the plot shows the trade-off controlled by λ: both very small or very large λ give less stable fits, small lambda underregularizes which causes overfitting and large lambda overregularizes which causes underfitting. The best value in this small experiment is in the middle of the tested range at roughly $\lambda = 0.1$.

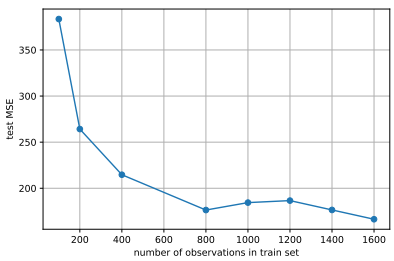


Figure 2.5: Test MSE as the number of training observations increases.

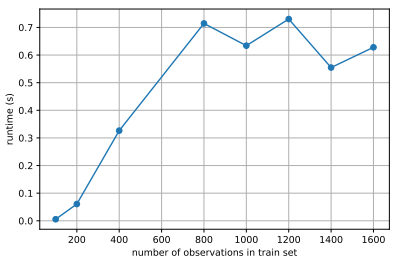


Figure 2.6: Runtime as the number of training observations increases.

Finally, we fix the test set and use increasingly larger subsets of the training set. This checks both the prediction error and the runtime as the number of training observations

grows.

Figures 2.5 and 2.6 show the basic trade off. More training observations can improve the test error but the runtime increases at the same time. This is the practical situation where Nyström becomes very useful: ideally we want to keep majority of the predictive information while avoiding the computational cost of dealing with the full kernel matrix.

This closes the motivation for Chapter 3. We have shown how KRR becomes a finite-dimensional kernel matrix problem, why the full matrix is expensive and why low-rank structure is worth exploiting. Chapter 3 therefore introduces the Nyström method as a way to approximate the positive semidefinite kernel matrix K using only selected columns.

3 Nyström Approximation

The Nyström approximation is a way to approximate a large positive semidefinite matrix by using only a smaller part of it. In this project the positive semidefinite matrix is the kernel matrix K, but it is useful to first describe the construction for a general positive semidefinite matrix A. The aim is to build a low rank approximation $\tilde{A}$ without using all of A in a full direct factorization.

3.1 General Nyström form

Following the general formulation in randomized numerical linear algebra, let

$$A \in \mathbb{R}^{n\times n}, \quad A \succeq 0,$$

and let $X \in \mathbb{R}^{n\times l}$ be a test matrix, the Nyström approximation of A with respect to X is

$$A\langle X\rangle = (AX)(X^TAX)^\dagger(AX)^T.$$

Here $A \succeq 0$ denotes that A is positive semidefinite, $(X^TAX)^\dagger$ denotes the Moore–Penrose pseudoinverse. The pseudoinverse generalizes the inverse to matrices that may be singular and it agrees with the usual inverse when X^TAX is invertible. The test matrix X is the thing we use to probe or sketch A by forming AX (Martinsson and Tropp, 2020). Later on when we sample columns this matrix becomes a sampling matrix that selects columns of the kernel matrix. The matrix AX contains the information we collect from A. The smaller matrix

$$X^TAX \in \mathbb{R}^{l\times l}$$

describes how the sampled directions interact with each other.

This formula shows why the approximation has rank at most l since the formula is a product involving the $n \times l$ matrix AX, so

$$\operatorname{rank}(A\langle X\rangle) \leq l.$$

Therefore, when $l \ll n$, the approximation is much cheaper to store and compute with than the full matrix A.

There is also a useful geometric representation of this formula. Since A is positive semidefinite, it has a positive semidefinite square root $A^{1/2}$ and $A = A^{1/2}A^{1/2}$. Therefore

$$AX = A^{1/2}(A^{1/2}X)$$

and

$$(AX)^T = (A^{1/2}X)^TA^{1/2}.$$

Also,

$$X^TAX = X^TA^{1/2}A^{1/2}X = (A^{1/2}X)^T(A^{1/2}X).$$

Substituting these into the Nyström approximation formula gives

$$A\langle X\rangle = A^{1/2}[(A^{1/2}X)((A^{1/2}X)^T(A^{1/2}X))^\dagger(A^{1/2}X)^T]A^{1/2}.$$

If we set

$$Y = A^{1/2}X.$$

Then

$$Y^TY = (A^{1/2}X)^T(A^{1/2}X) = X^TAX.$$

Then the expression inside the bracket becomes

$$Y(Y^TY)^\dagger Y^T.$$

We see that this is the standard formula for the orthogonal projector onto the column space of Y. Therefore, if we let P_Y denote this projector, then

$$A\langle X\rangle = A^{1/2}P_Y A^{1/2}.$$

Since $Y = A^{1/2}X$, we can also just write this as $P_{A^{1/2}X}$. Hence

$$A\langle X\rangle = A^{1/2}P_{A^{1/2}X}A^{1/2}.$$

This form explains the error formula. Since $A = A^{1/2}IA^{1/2}$, we have

$$A - A\langle X\rangle = A^{1/2}IA^{1/2} - A^{1/2}P_{A^{1/2}X}A^{1/2}.$$

Factoring out $A^{1/2}$ on both sides gives

$$A - A\langle X\rangle = A^{1/2}(I - P_{A^{1/2}X})A^{1/2}.$$

So the error is the part of A that is not captured by the subspace spanned by the columns of $A^{1/2}X$.

This has error as

$$A - A\langle X\rangle = A^{1/2}(I - P_{A^{1/2}X})A^{1/2}$$

which is the Schur complement viewpoint used by Martinsson and Tropp (Martinsson and Tropp, 2020). This means that the Nyström approximation keeps the part of A which is seen through the subspace generated by X. We now give a formal theorem of this.

Theorem 3.1. *Let $A \in \mathbb{R}^{n\times n}$ be positive semidefinite and let $X \in \mathbb{R}^{n\times l}$. Then*

$$A - A\langle X\rangle = A^{1/2}(I - P_{A^{1/2}X})A^{1/2}.$$

In particular,

$$A - A\langle X\rangle \succeq 0,$$

so

$$0 \preceq A\langle X\rangle \preceq A.$$

We have proved parts of this theorem above, to complete it we see that since $I - P_{A^{1/2}X}$ is an orthogonal projector, it is positive semidefinite. Thus

$$A - A\langle X\rangle \succeq 0.$$

This also gives

$$0 \preceq A\langle X\rangle \preceq A.$$

Theorem 3.1 identifies the error matrix of the Nyström approximation. The left hand side $A - A\langle X\rangle$ is exactly the part of A that is not captured by the approximation. The right hand side shows that this error is determined by the complementary projector $I - P_{A^{1/2}X}$. Since $P_{A^{1/2}X}$ projects onto the column space of $A^{1/2}X$, the matrix $I - P_{A^{1/2}X}$ projects onto complementary part of that space in the whole space. Therefore the theorem says that the Nyström error comes from the directions of A that are missed by the subspace generated by $A^{1/2}X$ (Martinsson and Tropp, 2020).

This shows why the choice of X matters. If the columns of $A^{1/2}X$ capture the important directions of A, then the missed part $A^{1/2}(I - P_{A^{1/2}X})A^{1/2}$ should be smaller. If they miss

important directions, then the approximation error can be large. In the column-sampling case, this means that the quality of the Nyström approximation depends on which columns are sampled. This motivates later sections to compare uniform sampling, leverage-score sampling and approximate leverage-score sampling.

The statement $0 \preceq A\langle X\rangle \preceq A$ says that the Nyström approximation does not create a matrix larger than A in the positive semidefinite order. This means that it does not invent or exaggerate in any directions in A and it approximates A from below. In the KRR setting this matters because replacing K by $\tilde{K}$ changes the regularized system from

$$(K + \lambda I)\alpha = y$$

to

$$(\tilde{K} + \lambda I)\tilde{\alpha} = y.$$

Since $\tilde{K} \preceq K$, the approximation changes both the estimator and the conditioning of the linear system. In directions where $\tilde{K}$ approximates K well, the solution should be close. In directions where $\tilde{K}$ loses a lot information, the λI regularization term dominates more. This is one reason why the number of sampled columns l, the sampling strategy and the regularization parameter λ have to be studied together.

3.2 Randomized Nyström approximation

We now discuss more on the test matrix X mentioned above that supposedly probes the important directions of A. The randomized Nyström approximation draws a random matrix $\Omega \in \mathbb{R}^{n\times l}$ and form $Y = A\Omega$. Then the Nyström approximation is

$$\hat{A} = Y(\Omega^T Y)^\dagger Y^T.$$

This is the randomized Nyström approximation described by Martinsson and Tropp (Martinsson and Tropp, 2020). The idea of a random test matrix was first introduced through the randomized rangefinder. In theory we can draw a random matrix Ω and form a sample matrix such as $Y = B\Omega$ for some matrix B, whose columns are used to capture an important subspace of the target matrix (Martinsson and Tropp, 2020, Section 11). A Gaussian test matrix is a standard theoretical choice where its entries are independent Gaussian random variables, so $B\Omega$ gives random linear combinations of the columns of B. This explains why randomization can be used to find a useful low-dimensional subspace. Nyström approximation therefore uses a similar idea but in this case the target matrix is positive semidefinite (Martinsson and Tropp, 2020, Section 14).

However, for kernel matrices, a dense Gaussian test matrix is usually not practical. Kernel matrices are expensive because even forming their entries requires $\mathcal{O}(n^2)$ kernel evaluations. In the coming sections we will see that the Nyström approximation is usually implemented by sampling columns of the kernel matrix. This is equivalent to using a column-selection matrix as the test matrix. Let $I = \{i_1, \ldots, i_l\} \subset \{1, \ldots, n\}$ be the selected column indices. Define the sampling matrix

$$Y = KS = K_{:,I}.$$

Therefore the Gaussian test matrix explains the general randomized sketching idea, while the column-selection matrix gives the practical Nyström approximation used in the experiments.

3.3 Nyström approximation and the KRR system

We now discuss the practical implementation of the Nyström approximation. Since K is symmetric positive semidefinite, it has an eigendecomposition

$$K = U\Lambda U^T.$$

Splitting the eigenvectors into essential and less important directions gives

$$U = \begin{bmatrix} U_1 & U_2 \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} \Lambda_1 & 0 \\ 0 & \Lambda_2 \end{bmatrix}.$$

Therefore

$$K = U_1 \Lambda_1 U_1^T + U_2 \Lambda_2 U_2^T.$$

If the eigenvalues in Λ_2 are small, then the second term contributes less to the matrix, and the ideal rank-l approximation is $K \approx U_1 \Lambda_1 U_1^T$. This motivates the approximation idea: instead of using all directions of K, we would like to keep only the important directions. We have seen this idea in the section on the general Nyström form, where the approximation error is determined by the directions that are missed by the subspace generated by X. In this case, the important directions are those corresponding to the eigenvectors in U_1.

However, computing U_1 and Λ_1 directly would require spectral information from the full kernel matrix. This is not practical if the goal is to avoid computing with all of K. Thus we construct a low rank approximation from sampled columns with $\tilde{K} = CW^{\dagger}C^T$. Here $C = K_{:,I}$ contains the sampled columns, and $W = K_{I,I}$ is the smaller intersection matrix. This is not the exact truncated eigendecomposition, but it is a column sampling way to approximate the important low rank part of K as we we will see in the next section (Musco and Musco, 2017).

3.4 Column sampling

We now look at column sampling. Let

$$I = \{i_1, \ldots, i_l\} \subset \{1, \ldots, n\}$$

be the selected column indices. Define the sampling matrix

$$S = \begin{bmatrix} e_{i_1} & e_{i_2} & \cdots & e_{i_l} \end{bmatrix} \in \mathbb{R}^{n \times l},$$

where e_i is the i-th standard basis vector. Taking $X = S$ in the general Nyström formula gives

$$\tilde{K} = K\langle S \rangle = (KS)(S^T K S)^{\dagger}(KS)^T.$$

Now define

$$C = KS = K_{:,I} \in \mathbb{R}^{n \times l}$$

and

$$W = S^T K S = K_{I,I} \in \mathbb{R}^{l \times l}.$$

Then the Nyström approximation becomes

$$\tilde{K} = CW^{\dagger}C^T.$$

Equivalently,

$$\tilde{K} = K_{:,I}(K_{I,I})^{\dagger} K_{I,:}.$$

This is the form used in this project. Instead of forming or storing all of K, the approximation is built from the sampled columns $K_{:,I}$ and the smaller intersection matrix $K_{I,I}$. The notation $K_{:,I}$ means that we keep all rows of K, but only the columns whose indices are in I. In other words,

$$K_{:,I} = \begin{bmatrix} K_{1,i_1} & K_{1,i_2} & \cdots & K_{1,i_l} \\ K_{2,i_1} & K_{2,i_2} & \cdots & K_{2,i_l} \\ \vdots & \vdots & & \vdots \\ K_{n,i_1} & K_{n,i_2} & \cdots & K_{n,i_l} \end{bmatrix}$$

Similarly, $K_{I,I}$ means that both the rows and columns only contain that of the selected index set I:

$$K_{I,I} = \begin{bmatrix} K_{i_1,i_1} & K_{i_1,i_2} & \cdots & K_{i_1,i_l} \\ K_{i_2,i_1} & K_{i_2,i_2} & \cdots & K_{i_2,i_l} \\ \vdots & \vdots & & \vdots \\ K_{i_l,i_1} & K_{i_l,i_2} & \cdots & K_{i_l,i_l} \end{bmatrix}$$

If W is invertible then the formula simplifies to

$$\tilde{K} = CW^{-1}C^T.$$

However in practice $W^\dagger$ is safer to write since W can be singular or nearly singular.

The column-sampling formula above also gives a simple Nyström algorithm. The steps are:

1. choose l landmark indices $I = \{i_1, \ldots, i_l\}$;
2. form matrix $C = K_{:,I}$ and $W = K_{I,I}$;
3. compute the pseudoinverse $W^\dagger$;
4. approximate K by $\tilde{K} = CW^\dagger C^T$.

The approximate KRR system obtained from the Nyström approximation is

$$(CW^{-1}C^T + \lambda I)\tilde{\alpha} = y.$$

If we first form the full matrix $CW^{-1}C^T$, then we again have an $n \times n$ matrix. This is useful in small experiments, but it does not use the full advantage of the Nyström representation. What we can do is to keep C and W, then reduce the solve to a smaller system.

Assume that W is invertible. Let

$$q = W^{-1}C^T\tilde{\alpha}.$$

Then

$$CW^{-1}C^T\tilde{\alpha} = Cq.$$

Therefore the approximate KRR system becomes

$$Cq + \lambda\tilde{\alpha} = y.$$

Solving this equation for $\tilde{\alpha}$ gives

$$\tilde{\alpha} = \frac{1}{\lambda}(y - Cq).$$

Substituting this into the definition of q gives

$$q = W^{-1}C^T\left(\frac{1}{\lambda}(y - Cq)\right).$$

Multiplying by λW gives

$$\lambda W q = C^T y - C^T C q.$$

Rearranging gives

$$(C^T C + \lambda W)q = C^T y.$$

This is only a $l \times l$ linear system. After solving for q, the approximate KRR coefficient vector is recovered by

$$\tilde{\alpha} = \frac{1}{\lambda}(y - Cq).$$

So the large system

$$(CW^{-1}C^T + \lambda I)\tilde{\alpha} = y$$

is replaced by the smaller system

$$(C^TC + \lambda W)q = C^Ty.$$

The full $n \times n$ matrix $CW^{-1}C^T$ is never formed explicitly. The main matrices used in the solve are $C \in \mathbb{R}^{n\times l}$, $W \in \mathbb{R}^{l\times l}$ and $C^TC \in \mathbb{R}^{l\times l}$.

The computational advantage is therefore as follows. Forming C^TC costs $O(nl^2)$ and solving the smaller system costs $O(l^3)$. When $l \ll n$, this is much cheaper than solving a dense $n \times n$ system, which costs about $O(n^3)$.

If W is singular, the inverse W^{-1} is replaced by the Moore–Penrose pseudoinverse $W^\dagger$. We factor $W^\dagger$ as $W^\dagger = BB^T$ and write

$$CW^\dagger C^T = CBB^TC^T = DD^T,$$

where $D = CB$. Then the approximate KRR system becomes

$$(DD^T + \lambda I)\tilde{\alpha} = y.$$

Now let $z = D^T\tilde{\alpha}$, then

$$DD^T\tilde{\alpha} = Dz,$$

so the system becomes

$$Dz + \lambda\tilde{\alpha} = y.$$

Solving for $\tilde{\alpha}$ gives

$$\tilde{\alpha} = \frac{1}{\lambda}(y - Dz).$$

Substituting this into the definition of z gives

$$z = D^T\left(\frac{1}{\lambda}(y - Dz)\right).$$

Multiplying by λ and rearranging gives

$$(D^TD + \lambda I)z = D^Ty.$$

After solving for z, the approximate KRR coefficient vector is recovered by

$$\tilde{\alpha} = \frac{1}{\lambda}(y - Dz).$$

Therefore, in the singular case, the large system

$$(CW^\dagger C^T + \lambda I)\tilde{\alpha} = y$$

is handled through the factorization

$$CW^\dagger C^T = DD^T$$

and reduced to the smaller system

$$(D^TD + \lambda I)z = D^Ty.$$

The full $n \times n$ matrix $CW^\dagger C^T$ is also never formed explicitly. The additional work is the factorization of $W^\dagger$, which only involves the small $l \times l$ matrix W.

3.4.1 Uniform sampling

The baseline method is uniform sampling. This means choosing l column indices from $\{1, \ldots, n\}$ uniformly at random. It is easy to implement and cheap because no importance scores have to be computed. Its weakness is that it treats all columns equally. If some columns are much more informative than others, uniform sampling may need a larger l to reach the same accuracy as a smarter strategy (Gittens and Mahoney, 2013, Lemma 8).

Before stating the theorem for the uniform sampling error bound, we define the leverage scores and coherence used in the bound.

Let $K \in \mathbb{R}^{n \times n}$ be a positive semidefinite kernel matrix with eigendecomposition

$$K = U \Lambda U^T.$$

Let $U_k \in \mathbb{R}^{n \times k}$ contain the eigenvectors corresponding to the k largest eigenvalues of K. Following Gittens and Mahoney, the *statistical leverage score* of row or column i relative to the best rank-k approximation is defined as

$$s_i = \|U_k(i, :)\|_2^2.$$

These scores measure how much the i-th coordinate contributes to the leading k-dimensional eigenspace. Since U_k has orthonormal columns, the scores satisfy

$$\sum_{i=1}^{n} s_i = k.$$

The *coherence* is defined as

$$\mu = \frac{n}{k} \max_{1 \le i \le n} s_i.$$

Thus μ is the scaled version of the largest leverage score (Gittens and Mahoney, 2013).

We also introduce the idea of best rank-k approximation. From the same eigendecomposition, the best rank-k approximation is

$$K_k = U_k \Lambda_k U_k^T,$$

where U_k contains the eigenvectors corresponding to the k largest eigenvalues as seen before. This matrix is the best rank-k approximation to K in spectral norm. In the Nyström settings, k is therefore the optimal rank we compare against while l is the number of columns actually sampled. In practice l should be larger than k, since random column sampling needs to oversample to capture the important k-dimensional subspace with high probability.

The leverage score measures how concentrated the leading eigenspace is in a few rows. If the scores are all similar, then μ is not large and uniform sampling has a better chance of working. If a few scores are much larger than the rest, then uniform sampling can easily miss important columns. Thus uniform sampling can be reasonable in some datasets where the leverage scores are fairly uniform. In other datasets where the leverage scores are highly nonuniform, uniform sampling can be much worse (Gittens and Mahoney, 2013, Lemma 8). This is why uniform sampling is a natural baseline method but not necessarily the best method.

Finally we have the theorem, and it uses two probability parameters. The failure probability δ controls how likely it is that the random sampling guarantee fails, so a smaller δ means higher confidence. The accuracy factor ϵ controls the sampling condition, a smaller ϵ represents a stronger guarantee and therefore require more columns to be sampled(Gittens and Mahoney, 2013, Lemma 8).

Theorem 3.2 (Uniform sampling)**.** *Let $K \in \mathbb{R}^{n\times n}$ be positive semidefinite, and let K_k be its best rank-k approximation. Fix a failure probability δ and an accuracy factor ϵ both in $(0,1)$. Suppose l columns are sampled uniformly at random, then*

$$C = KS = K_{:,I}, \qquad W = S^T KS = K_{I,I}.$$

If

$$l \geq 2\mu\epsilon^{-2}k\ln(k/\delta),$$

then, with probability at least $1-3\delta$, the Nyström approximation $\tilde{K} = CW^{\dagger}C^T$ satisfies the spectral norm bound

$$\|K - \tilde{K}\|_2 \leq \left(1 + \frac{n}{(1-\epsilon)l}\right)\|K - K_k\|_2.$$

This bound for l fits the intuition as it shows us when μ is large we need to sample more columns. This is exactly the point that uniform sampling does not know which columns are important so when the coherence is large it needs more columns to capture the essential structure.

In the first experiments we test uniform sampling. For each l, the approximation quality can be measured directly by a relative forward error, for example

$$\frac{\|K - \tilde{K}\|_F}{\|K\|_F}.$$

It is also important to measure the effect on prediction. So the approximation is also judged by what happens when $\tilde{K}$ is used in place of K in the regularized system, this is done by solving the system with $\tilde{K}$ and then measuring the test error of the result.

3.4.2 Leverage score sampling

Uniform sampling is cheap but it ignores the structure of K. Leverage score sampling is the opposite extreme, it tries to sample columns according to how important they are in the leading eigenspace.

The full leverage score version is the most informed method. It uses all of K first:

1. construct the full kernel matrix K;
2. compute the leading eigenvectors U_k where U_k comes from the eigendecomposition $K = U\Lambda U^T$ as before;
3. compute the squared row norms $s_i = \|U_k(i,:)\|_2^2$;
4. compute probabilities $p_i = \frac{s_i}{\sum_j s_j}$;
5. sample columns according to these probabilities.

This method is expensive because it requires the full matrix and the leading eigenvectors. But it is still useful as a reference method. It says what the informed sampling distribution would be if the important eigenspace was already known.

Theorem 3.3 (Leverage score sampling)**.** *Let $K \in \mathbb{R}^{n\times n}$ be positive semidefinite and let U_k contain the leading k eigenvectors. Suppose columns are sampled with probabilities satisfying*

$$p_i \geq \frac{\beta}{k}\|U_k(i,:)\|_2^2$$

for some $\beta \in (0, 1]$. *If the number of sampled columns has order*

$$l = \mathcal{O}\left(\frac{k \log(k/\beta)}{\beta \epsilon^2}\right),$$

then the Nyström approximation satisfies

$$\|K - \tilde{K}\|_2 \leq \|K - K_k\|_2 + \epsilon^2 \|K - K_k\|_*,$$

where ϵ *is a fixed approximation factor and* $\|\cdot\|_*$ *denotes the nuclear norm which is the sum of a matrix's singular values.*

This theorem can be compared with the uniform sampling theorem. In the uniform sampling theorem, the required number of columns depends on the coherence μ, because uniform sampling does not know which columns are important. In the leverage score result, the sampling probabilities depend on the row norm scores $\|U_k(i,:)\|_2^2$, so the sample-size condition no longer has the same coherence factor (Gittens and Mahoney, 2013).

The term $\|K - K_k\|_2$ is the spectral norm error of the best rank-k approximation, so this is the error that's unavoidable if only rank k information is kept. The theorem says that leverage score sampling can get close to this best rank-k error, with an additional tail term $\epsilon^2 \|K - K_k\|_*$. This extra term depends on the nuclear norm of the remaining tail. Therefore, if the eigenvalues after the first k are small, the additive term is also small.

4 Approximate leverage score sampling

The full leverage score method is too expensive if the whole reason for Nyström is to avoid working with all of K. The middle method is therefore to look at a small part of K first and use it to estimate which rows or columns look important.

Here is a simple version of this idea:

1. sample 5 columns uniformly;
2. form the small sampled matrix $K_s \in \mathbb{R}^{n \times 5}$;
3. compute an orthonormal basis U_s for the column space of K_s;
4. compute approximate scores as before.
5. sample 95 more columns (total 100 columns) according to these approximate scores.

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